a Boson

columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, J_1^2

g I comm.	g I anti.	g -I comm.	g -I anti.	-g I comm.	-g I anti.	-g -I comm.	-g -I anti.
g, I	B01 d=4 G; +Θ3 cone: Lemma 2	B02 d=4 C(iJ) strict cone	B03 d=2 G; +Θ3 cone: Lemma 1	B04 d=4 Θ(+,-)+RP; p=1 cone: Lemma 1	B05 d=2 C(J) strict cone	B06 d=4 G; +Θ3 cone: Lemma 2	B07 d=4 Θ(+,-); p=1 cone: Lemma 1	B08 d=4 G; +Θ3 cone: Lemma 1
g, -I	B09 d=2 G; +Θ3 cone: Lemma 2	B10 d=4 C(iJ) strict cone	B11 d=4 G; +Θ3 cone: Lemma 1	B12 d=4 Θ(-,-); p=0 cone: Lemma 1	B13 d=4 C(J) strict cone	B14 d=2 G; +Θ3 cone: Lemma 2	B15 d=4 Θ(-,-); p=1 cone: Lemma 1	B16 d=4 G; +Θ3 cone: Lemma 1
-g, I	B17 d=2 C(iJ) strict cone	B18 d=4 G; +Θ3 cone: Lemma 2	B19 d=4 Θ(+,-); p=1 cone: Lemma 1	B20 d=2 G; +Θ3 cone: Lemma 1	B21 d=4 G; +Θ3 cone: Lemma 2	B22 d=2 C(J) strict cone	B23 d=4 G; +Θ3 cone: Lemma 1	B24 d=4 Θ(+,-); p=1 cone: Lemma 1
-g, -I	B25 d=4 C(iJ) strict cone	B26 d=4 G; +Θ3 cone: Lemma 2	B27 d=4 Θ(-,-); p=1 cone: Lemma 1	B28 d=4 G; +Θ3 cone: Lemma 1	B29 d=4 G; +Θ3 cone: Lemma 2	B30 d=4 C(J) strict cone	B31 d=8 G; +Θ3 cone: Lemma 1	B32 d=4 Θ(-,-); p=0 cone: Lemma 1

b Fermion

columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, J_1^2

g I comm.	g I anti.	g -I comm.	g -I anti.	-g I comm.	-g I anti.	-g -I comm.	-g -I anti.
g, I	F01 d=2 SO / split-O cone: Lemma 1	F02 d=4 G; +Θ3 cone: Lemma 1	F03 d=4 G; +Θ3 cone: Lemma 2	F04 d=4 C(iJ)+Θ strict cone	F05 d=2 G; +Θ3 cone: Lemma 1	F06 d=2 G; +Θ3 cone: Lemma 1	F07 d=2 C(J)+Θ strict cone	F08 d=2 G; +Θ3 cone: Lemma 2
g, -I	F09 d=4 G; +Θ3 cone: Lemma 1	F10 d=4 Θ(+,-) cone: Lemma 1	F11 d=4 G; +Θ3 cone: Lemma 2	F12 d=8 C(iJ)+Θ strict cone	F13 d=4 Θ(+,-) cone: Lemma 1	F14 d=4 G; +Θ3 cone: Lemma 1	F15 d=4 C(J)+Θ strict cone	F16 d=4 G; +Θ3 cone: Lemma 2
-g, I	F17 d=2 G; +Θ3 cone: Lemma 1	F18 d=2 G; +Θ3 cone: Lemma 1	F19 d=4 C(iJ)+Θ strict cone	F20 d=4 G; +Θ3 cone: Lemma 2	F21 d=4 G; +Θ3 cone: Lemma 1	F22 d=4 G; +Θ3 cone: Lemma 1	F23 d=4 G; +Θ3 cone: Lemma 2	F24 d=2 C(J)+Θ strict cone
-g, -I	F25 d=2 Θ(+,-) cone: Lemma 1	F26 d=2 G; +Θ3 cone: Lemma 1	F27 d=4 C(iJ)+Θ strict cone	F28 d=4 G; +Θ3 cone: Lemma 2	F29 d=4 G; +Θ3 cone: Lemma 1	F30 d=2 Θ(+,-) cone: Lemma 1	F31 d=4 G; +Θ3 cone: Lemma 2	F32 d=4 C(J)+Θ strict cone

strict cone, $\eta = J_2 g$
strict cone, $\eta = iJ_2 g$
bosonic physical anti-linear pair (p = low-d physical orbits)
bosonic reflected-factor class plus RP condition
fermionic anti-linear, compact-SO, or split-O theorem
group only; $\mathrm{Tr} H = 0$
group only; J_1 flips H